

SEBIN THOMAS

LEAD SOFTWARE ENGINEER



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ABOUT ME

Software Engineer with 6+ years of experience developing firmware and HMI applications for automotive products. Proficient in C and C++ programming on embedded platforms (Linux, Android, FreeRTOS), with hands-on expertise in Bluetooth/BLE, CAN, UART, SPI, and I²C communication protocols. Experienced in using development tools such as MCUXpresso IDE, Code Composer Studio, Kanzi Studio, and CANoe, along with unit documentation tools (Doxygen). Known for adhering to MISRA coding standards and ASPICE processes, with strong teamwork and communication skills to deliver high-quality software solutions.

EXPERIENCE

Johnsoft Technologies LLP - Lead Software Engineer

June 2019

Instrument Cluster Application - QNX Middleware

- Middleware application development for the infotainment features of the instrument cluster
- Data is received through the gRPC and send the client using PPS interface
- Testing of the infotainment features - Media, Phone, Navigation etc.
- Prepared technical documentation using ASPICE standard automotive documentation.
- Technologies : C++, Visual Studio, QNX, gRPC, PPS

Instrument Cluster Application - Kanzi Bussiness Logic (QNX)

- Implemented HMI logic and application features using C++.
- Executed functional and integration testing to validate system behavior using CANoe tool.
- Prepared technical documentation using ASPICE standard automotive documentation.
- Technologies : C++, Visual Studio, CANoe, QNX

Instrument Cluster Application - UI (Kanzi)

- Implemented UI based on the figma.
- Executed functional and integration testing to validate UI design using Kanzi environment.
- Prepared technical documentation using ASPICE standard automotive documentation.
- Technologies: C++, Visual Studio, Kanzi Studio,

Bluetooth KeyFob Middleware on TI SoC (Vehicle door control and localization)

- Software development for KeyFob application (remote door lock/unlock, localization etc.) on over BLE communication, TI SoC platform, with TI RTOS.
- Testing of SW interface layer (middleware) for BLE stack – connection, pairing, registration SW for KeyFob message handling and interface to CAN module
- Performed unit testing using GTest and GMock to ensure code reliability and coverage.
- Technologies : C, Code Compose Studio, XDS110 debugger, TI RTOS, CC2340

Bluetooth KeyFob Middleware on NXP SoC (Vehicle door control and localization)

- Software development for KeyFob application (remote door lock/unlock, localization etc.) on over BLE communication, on NXP SoC platforms, with NXP RTOS.
- Testing of SW interface layer (middleware) for BLE stack – connection, pairing, registration, encryption etc., both central and peripheral SW for KeyFob message handling and interface to CAN module
- repared technical documentation using doxygen.
- Technologies : C, MCUXpresso IDE, XDS110 debugger, NXP RTOS, KW45

Z-wave Adapter for Smart Home Controllers

- Software development for smart Home Controlllers on over Z-Wave Protocol,
- This application is a middleware that interface with MQTT and Z-Wave
- Responsibilities: Design, Implementation, Testing
- Technologies : C++, ZWave, ZIP-Gateway

MQTT Router for Smart Home Controllers

- Software development for smart Home Controlllers on over MQTT Protocol,
- This application is a middleware that interface with MQTT and Z-Wave
- Responsibilities: Design, Implementation, Testing
- Technologies : C++, MQTT

Apple CarPlay MiddleWare – In Vehicle Infotainment

- Software development for apple carplay middleware for an android-based IVI system,
- This middleware will interface between apple CarPlay chip on the IVI and IVI
- Responsibilities: Design, Implementation, Testing
- Technologies : C++, Visual Studio, AOSP

Android Native Development – Bluetooth

- Resolving the Issues on the Android Native Bluetooth Framework
- Customizing based on the requirements
- Worked on the A2DP, AVRCP, HFP etc
- Technologies : Java, AOSP

Android WebSocket Middleware

- Software development for WebSocket Based Communication middleware module
- Helps to Controls the supported features remotely
- Technologies: Java, AOSP

EDUCATION

B-TECH - CSE

Indira Gandhi College of Engineering and Technologies, Kothamangalam, Kerala (2015 – 2019)

APJ Abdul Kalam Technological University

SKILLS

- Programming Languages: C, C++, java, Kotlin
- Operating Systems: Linux (Ubuntu), Windows, Android
- Development Tools: MCUXpresso IDE, VS Code, Code Composer Studio IDE, Kanzi Studio
- Boards worked: Texas Instruments (CC26X2, CC2340), NXP (KW38, KW45)
- Documentation Tool: Doxygen
- Hands on experience in BLE sniffing tools, CANoe tool, Renesas debugger tool.
- Hands on experience in RTOS, FreeRtos, QNX,
- Tools for Requirements, Integration Test, System Test: Polarion
- Version Control: Git/Gerrit
- Task Management: Jira
- Hands on experience in communication protocols – BT/BLE, UART, SPI, I2C
- Experience in ASPICE SWE.1 to SWE.6
- Good communication skills and a good team player

ADDITIONAL INFORMATION

Strengths

- Strong creative & innovative ability.
- Good reasoning power.
- Determined to learn with practical approach.
- Enthusiastic and can produce results under the deadline constraints.
- Languages Known : English, Malayalam, Tamil

DECLARATION

I hereby declare that the above furnished particulars are true to the best of my knowledge and belief.

SEBIN THOMAS